1.1 Introduction

The use of databases and database systems is crucial to everyday life. We have been using non-computerized databases for a very long time; for example, when we look up a word in a dictionary or look up a friend's phone number in a phone book. Computerised databases are used to store, manage, and manipulate databases. In our daily lives, we interact with computerised databases when we reserve train and movie tickets, look up a book in a library, get information about our salaries, check the balance of our accounts at ATMs, and get price lists when we shop at cash and carry stores. This list can be quite long. Recently, databases have also been used to store highly data-intensive items like images (http://images.google.co.in), video clips (http://youtube.com), and photographs (http://flickr.com).

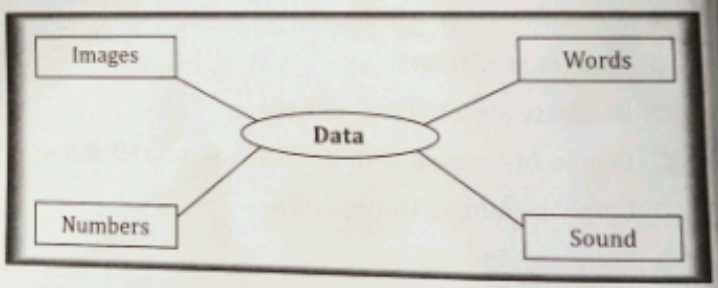
The database system makes most of the operations automated. A very good example for this is the billing system used for the items purchased in a super market. Obviously this is done with the help of a database application package. Inventory systems used in a drug store or in a manufacturing industry are some more examples of database.

1.2 Basic Definitions and Concepts

All of us are familiar with the term data. In fact, unknowingly we come across data in our day to day life every day. The age of a person, price of potato, number of students in a school, pin code of a city, etc. are some examples of data.

1.2.1 Data

Data is a raw and unorganized fact that is required to be processed to make it meaningful. Data can be simple at the same time unorganized unless it is organized. Generally, data comprises facts, observations, perceptions, numbers, characters, symbols, images, etc.



Definition: Data

Data is the raw material from which useful information is derived. The word data is the plural of Datum. Data is commonly used in both singular and plural forms. It is defined as raw facts or observations. It takes variety of forms, including numeric data, text and voice and images. Data is a collection of facts, which is unorganized but can be made organized into useful information. The term Data and Information come across in our daily life and are often interchanged.

We may be in a position to tell accurately the age, height, complexion, income, educational qualification, residential address, etc. of our close friends. But it is too difficult for us to memorize all these information for a large number of individuals. To deal with such problems we need a database.

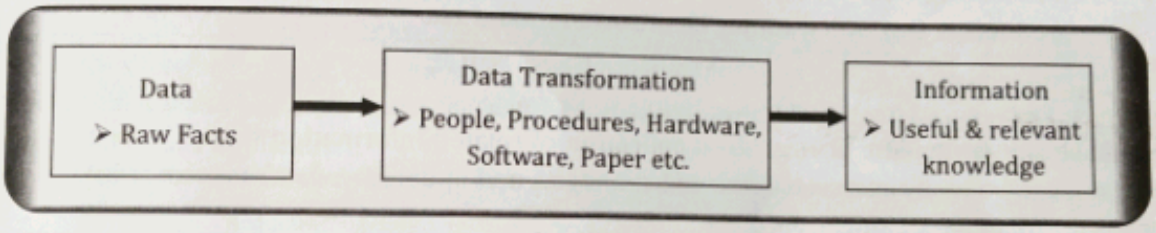
1.2.2 Information

Information can be explained as any kind of understanding or knowledge that can be exchanged with people. It can be about facts, things, concepts, or anything relevant to the topic concerned.

Definition : Information

It is a processed data (or) collection of data which is having perfect meaning is called information.

Example: The temperature of the days is ***data***. When this data is to be collected, a system or person monitors the daily temperatures and records it. Finally when it is to be converted into meaningful ***information***, the patterns in the temperatures are analysed and a conclusion about the temperature is arrived at. So information obtained is a result of analysis, communication, or investigation.



Information

Differences between Data and Information

Data	Information
Data is raw, unorganized facts that need to be processed. Data can be something simple and seemingly random and useless until it is organized.	When data is processed, organized, structured or presented in a given context so as to make it useful, it is called Information
Data is atomic level pieces of information	Information is a collection of data
Data does not help in decision taking	Information helps in decision taking
Data is the plural of the Latin word datum. It can mean 'to give'.	The word information was derived from Latin. The verb from which it is derived is inform are, which means 'to instruct'. It also means giving form to an idea or fact.
Example: Each student's test score is one piece of data	**Example:** The class' average score or the school's average score is the information that can be concluded from the given data.

1.2.3 Data Base

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. Databases make data management easy.

It is an organized collection, because in a database, all data is described and associated with other data. All information in a database should be related as well; For example, a database that contains information about students should not hold information about company stock prices.

Database contains data based on the kind of application for which it is required.

Examples of Database:

- **Online Telephone Directory :** An online telephone directory uses a database to store data of people, phone numbers, and other contact details.
- **Facebook :** Facebook needs to store, manipulate, and present data related to members, their friends, member activities, messages, advertisements, and a lot more.
- **Airline Database:** An airline database may contain data about the airplane, the routes, airline reservation, airline schedules etc.
- **Dictionary :** A database of words organized alphabetically along with their meaning.
- **College Database :** A college database may contain data about the students, faculty, administrative staff, courses, results etc.,
- **Manufacturing Database:** A database for manufacturing application may contain data about the production, inventory, supply chain, orders, sales etc.

Definition : Database

A database is a systematic or organized collection of related information that is stored in such a way that it can be easily accessed, retrieved, managed, and updated. A database stores information, which is useful to an organization.

1.3 Why Database?

A database is a collection of organized data, usually stored in electronic form. A database is typically designed so that it is easy to store and access information. A good database is crucial to any company or organisation. There are various reasons to use a database. The important ones are listed below.

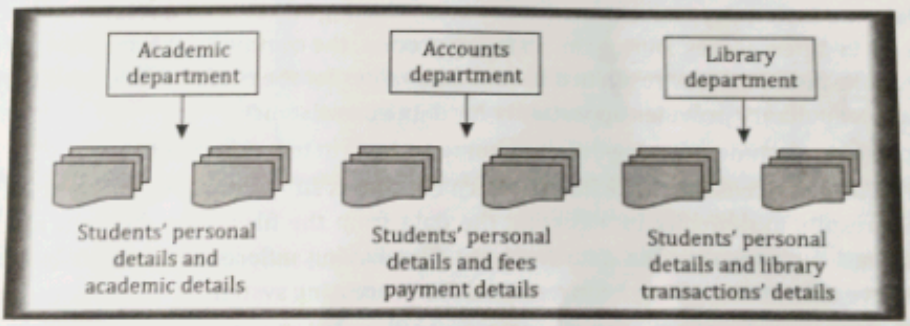
1. **Manages large amounts of data :** A database stores and manages a large amount of data on a daily basis.
2. **Database Stores Information Easily :** The database helps manage information more effectively. **It also allows to add, delete, or modify information quickly and easily.**
3. **Accurate : It helps to keep data up-to-date and accurate.**
4. **Database Is Reliable :** Databases store data consistently and reliably. This helps to prevent the data from becoming corrupted. If the data is stored in a database, it can be easily recovered, even if it is corrupted. However, if the information is stored on paper, it may be difficult or impossible to retrieve the data if it becomes damaged or lost.
5. **Database Is Easy To Use :** Databases are easy to use. They can be accessed from any computer with an internet connection. This allows to quickly and easily get the required information.
6. **Database Ensures Data Security :** The data is secure when it's kept in a database. Only authorized individuals can access the information, which prevents unauthorized users from gaining access to the sensitive data. Databases use a variety of security mechanisms to protect data.
7. **Data integrity:** The data integrity is ensured in databases by using various constraints for data. Data integrity in databases makes sure that the data is accurate and consistent in a database.

1.4 Traditional File Based Approach

A file system is a method for storing and organizing computer files and the data they contain to make it easy to find and access them. File systems may use a storage device such as a hard disk or CD-ROM and involve maintaining the physical location of the files.

The term **'file-based approach'** refers to the situation where data is stored in one or more separate computer files defined and managed by different application programs. Typically, for example, a student information system would include files like student profile, student course, student result, student fees etc. Computer programs or computer applications access the stored files to perform the various tasks required by the business. The computer applications are written to access the data in the files and hence applications are built on top of the file system.

In the file-based approach of data processing, each department has its own files, which are specifically designed for that department's applications. The below figure shows three departments academics, accounts, and library, Each of these departments maintains a file containing students' personal details in addition to the file that is necessary for its own application, Therefore, each application has a separate master file and its own set of personal files, The following are the various types of files maintained for each application.



File-oriented Approach

1.4.1 Advantages of File Based Approach

The file based approach is simple and less expensive. It fits the needs of many small businesses. The customization is easy and efficient. The other advantages of file based approach are listed below.

Advantages of File Based Approach

1. **Backup:** It is possible to take faster and automatic back-up of database stored in files of computer-based systems.
2. **Compactness:** It is possible to store data compactly.
3. **Data Retrieval:** Computer-based systems provide enhanced data retrieval techniques to retrieve data stored in files in easy and efficient way.
4. **Editing:** It is easy to edit any information stored in computers in form of files. Specific application programs or editing software can be used for this purpose.

5. **Remote Access:** In computer-based systems,it is possible to access data remotely. so,to access data it is not necessary for a user to remain present at location where these data are kept.
6. **Sharing:** Data stored in files of computer-based systems can be shared among multiple users at a same time.

1.4.2 Disadvantages of File-Based Approach

There are many drawbacks of using the file system. The important drawbacks of file based approach are:

Disadvantages of File Based Approach

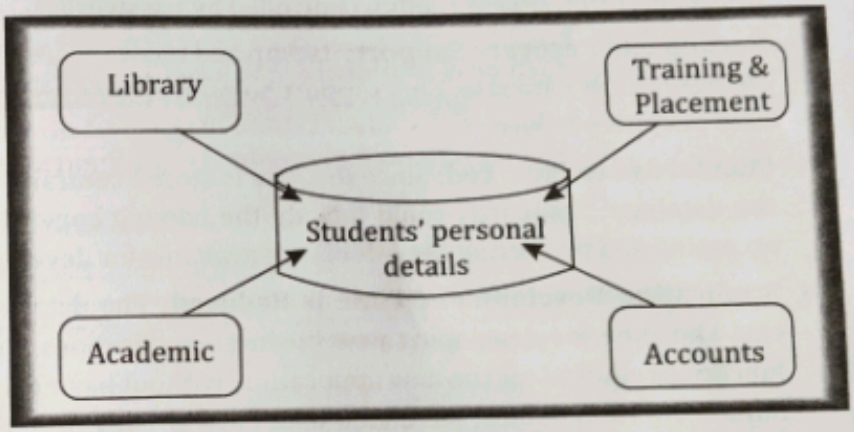
1. **Data Redundancy :** Data redundancy means storing the same data at multiple locations. In an application, a file may have fields that are common to more than one file. The data for these common fields is thus replicated in all the files having these fields. This results in data redundancy, as more than one file has the same data values stored in them.
   Example : student_name and student_course may be stored in two files—"student profile" and "student fees".
2. **Data Inconsistency :** Data inconsistency means having different data values for the common fields in different files. During the updating process, the common fields may not get updated in all the files. This may result in different data values for the common fields in different files. Data redundancy provides opportunity for data inconsistency.
   Example : A student having different home address in two different files.
3. **Difficulty in Accessing :** The files in which the data is stored can have different file formats. This results in difficulty in accessing the data from the files since different methods are required for accessing the data from the files having different formats It is not easy to retrieve information using a conventional file processing system.
4. **Data Isolation :** Data are scattered in various files, and the files may be in different format, writing new application program to retrieve data is difficult.
5. **Integrity Problems :** The data stored in the file must satisfy certain types of consistency constrains. When constraints have to be added, it is difficult to change the programs to enforce them. The problem is even more complex when constraints involve several data items from different files.
   Example : The balance field Value must be grater than 5000. We have to handle this through program code.
6. **Atomicity Problem : :** In case of a failure, data must be restored to the correct state that existed prior to the failure, It is difficult to accomplish this in a file-oriented system.
   Example : Transferring ₹100 from Account A to Account B. If a failure occurs during execution there could be situation like ₹100 is deducted from Account A and not credited in Account B.
7. **Concurrent Access Anomalies :** The files can be accessed concurrently by multiple users. Uncontrolled concurrent access may lead to inconsistency and security problems. For example, two users may try to update the data in a file at the same time.

8. **Security Problems :** Not every user should be able to access all the data For example, a librarian should not be allowed to access the salary details of teachers, However, enforcing security constraints in a fileoriented system is difficult.
9. **Poor Data Modeling of Real World :** The file based system is not able to represent the complex data and inter-file relationships, which results poor data modeling properties.

1.5 Database Approach

Database approach provides solutions for handling the problems of the file system approach. The emergence of database approach has resulted in a paradigm shift, from each application defining and maintaining its own data (file-oriented approach)—to the data being defined and administered centrally (database approach). In the database approach, data is defined and stored centrally.

A database is a collection of related data that is organized in a way that allows users to easily access, update, and maintain the data. It stores non-redundant data that can be shared by different application systems. A non-redundant data means that only a single copy of a data exists in the entire system. Non-redundancy ensures data consistency.



Database Approach

The below figure shows the database approach in which all the programs access a single copy of a data in the database.

1.5.1 Benefits of Database over File System (Advantages of Databases / DBMS)

The main characteristics and benefits of the database approach are defined as follows:

- **Data Redundancy is Minimized:** Database system keeps data at one place in the database. The data is integrated into a single, logical structure. Different applications refer to the data from the centrally controlled location. The storage of the data in central location minimizes data redundancy.
- **Data Inconsistency is Reduced:** Minimizing data redundancy using database system reduces data inconsistency too. For example, students' home addresses are stored at a single location and get updated centrally.
- **Data is Shared:** Data sharing means sharing the same data among more than one user. Each user has access to the same data, though they may use it for different purposes. The database is designed to support shared data. Authorized users are permitted to use the data from the database. Users are provided with views of the data to facilitate its use. For example, the students' home addresses stored in the database which is shared by student profile system and library system.

- **Data Independence:** It is the separation of data description (metadata) from the application programs that use the data. In the database approach, data descriptions are stored in a central location called the data dictionary. This property allows an organization's data to change and evolve (within limits) without changing the application programs that process the data.
- **Data Integrity is Maintained:** Stored data is changed frequently for variety of reasons such as adding new data item types, and changing the data formats. The integrity and consistency of the database are protected using constraints on values that data items can have. Data constraint definitions are maintained in the data dictionary.
- **Data Security is Improved:** The database is a valuable resource that needs protection. The data base is kept secure by limiting access to the database by authorized personnel. Authorized users are generally restricted to the particular data they can access, and whether they can update it or not. Access is often controlled by passwords.
- **Backup and Recovery Support:** Backup and recovery are supported by the software that logs changes to the database. This support helps in recovering the current state of the database in case of system failure.
- **Standards are Enforced:** Since the data is stored centrally, it is easy to enforce standards on the database. Standards could include the naming conventions, and standard for updating accessing and protecting data. Tools are available for developing and enforcing standards.
- **Application Development Time is Reduced:** The database approach greatly reduces the cost and time for developing new business applications. Programmer can focus on specific functions required for the new application, without having to worry about design, or low-level implementation details; as related data have already been designed and implemented. Tools for the generation of forms and reports are also available.
- **Representing Complex Relationships Among Data:** A DBMS have the capability to represent complex relationships and to retrieve related data efficiently.

1.5.2 Drawbacks of Database over File System (Disadvantages of Databases / DBMS)

Database approach is quite useful compared to the file based approach. However, it does have some disadvantages. Some of those are as follows.

- **Expensive :** Creating and managing a database is quite costly. High cost software and hardware is required for the database. Also highly trained staff is required to handle the database and it also needs continuous maintenance. All of these ends up making a database quite a costly venture.
- **High Complexity :** A database approach is quite complex as it involves creating, modifying and editing a database. Consequently, the people who handle a database or work with it need to be skilled otherwise valuable data can be lost.
- **Database Failure :** All the relevant data for any company is stored in a database. A database failure can lead to loss or corruption of very important data.
- **High Maintenence :** A database contains a large amount of data, especially for bigger organisations. It is more difficult to handle and maintain large databases. It is also more complex to ensure data consistency and user authentication across big databases.

- **Upgradation Costs :** Often new functionalities are added to the database. This leads to database upgradations. All of these upgradations cost a lot of money.
- **Cost of Data Conversion :** If the database is changed or modified in some manner, all the data needs to be converted to the new form. This cost may even exceed the database creation and management costs sometimes. This is the reason most organisations prefer to work on their old databases rather than upgrade to new ones.

1.6 File Based Approach Vs Database Approach

File Based Approach	Database Approach
Used to manage and organise the files stored in the hard disk of the computer.	A software is used to store and retrieve the user's data.
High data inconsistency.	Maintains data consistency.
Strcuture is simple.	Structure is complex.
Data sharing is hard.	Data sharing is easy.
High data redundancy.	Low redundancy.
Less Secure.	High Secure.
No Backup and Recovery Process	Supports both Backup and Recovery.
Less Expensive.	High Expensive.
No Data Independence	Data Independence exists,
Difficult to store and retrieve data.	Easy to store and retrieve data.
Integrity Constraints are difficult to implement.	Integrity Constraints are easy to apply.
There is no efficient query processing in the file system.	Efficient query processing is available.
It give details of storage and representation of data.	It hides the internal details of Database
Example: COBOL, C++	Example: Oracle, SQL Server

1.7 Characteristics and Purpose of Database Approach

The following are the different characteristics of database approach:

1. Self-describing nature of a database system
2. Insulation between programs and data
3. Data Abstraction
4. Support of multiple views of the data
5. Sharing of data and multiple-user transaction processing

1.7.1 Self-Describing Nature of a Database System

A fundamental characteristic of the database approach is that the database system contains not only the database structure and constraints and also contains the definition which is stored in the system catalog. A DBMS catalog stores the description of a particular database.

Example: Data structures, types and constraints

This description of the catalog is meta-data, this allows the DBMS software to work with different database application. **Meta-data** (i.e., data about data) is stored in the so-called **system catalog** which contains a description of the structure of each file, the type and storage format of each field, and the various constraints on the data

1.7.2 Insulation between Programs and Data

In traditional file processing, the structures of data files is embedded in the accessing programs, so many changes to the structure of a file may require changing all programs that access this file. By contrast, **DBMS** access programs do not require such changes in most cases. The structure of data files is stored in the DBMS catalog separately from the access programs. We call this property as **program data independence**: which allows changing data structures and storage organization without having to change the DBMS access programs.

1.7.3 Data Abstraction

The characteristic that allows program data independence and program operation independence is called **data abstraction**. A data model is a type of data abstraction that is used to hide storage details and present the users with a conceptual view of the database.

Support of multiple views of the data : Each user may see a different view of the database, which describes only the data of interest to that user. A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored. Different users (example in different departments of an organization) have different "views" or perspectives on the database.

Example: Consider library database system. This database may be used by different users like Researchers, Teachers, Students etc. For different - perspectives. Therefore all the users are using the same database from different point of view. This characteristic is called **support of multiple views of data** (This is an example of a **subset** view.)

1.7.4 Sharing of Data and Multi-user Transaction Processing

It means allowing a set of concurrent users to retrieve from and to update the database.

Concurrency control within the DBMS guarantees that each transaction is correctly executed (or) aborted. Recovery subsystem ensures each completed transaction has its effect permanently recorded in the database. OLTP (online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second. Applications such as **airline reservation systems** are known as **online transaction processing** applications

1.8 Data Base Management System (DBMS)

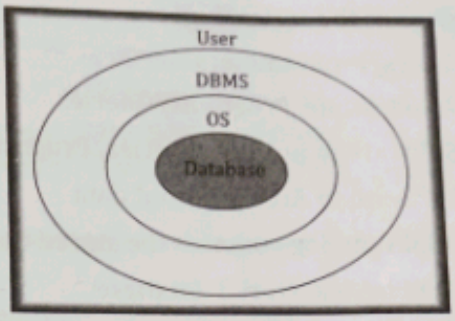
A database management system is software that manages a database by storing, retrieving, and manipulating the data from a database in efficient manner.

A database management system (DBMS) enables users to create and maintain a database. The DBMS is a general-purpose software system that facilitates the processes of defining, constructing,

manipulating, and sharing databases among various users and applications. It also helps to control access to the database.

Examples of DBMS Softwares: Oracle Database, Microsoft SQL Server, MySQL, IBM DB2, PostgreSQL.

Database Systesm : The database and the DBMS software together is a **Database system.**

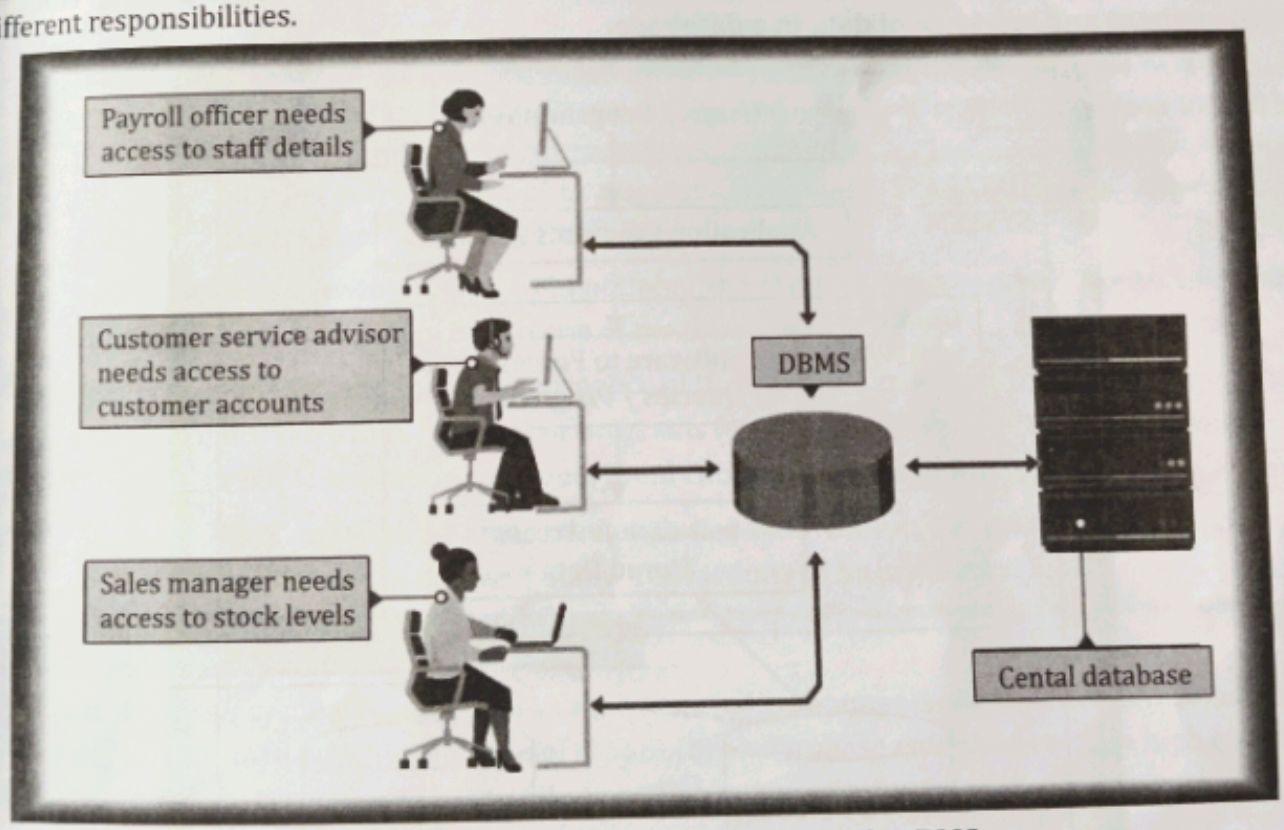


Database System

Definition: Database Management System (DBMS)

A database management system is software that manages a database by storing, retrieving, and manipulating the data from a database in efficient manner.

The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient. A database in a DBMS could be viewed by lots of different people with different responsibilities.



Empolyees are Accessing Data through DBMS

1.9 Simplified Database System

A database system is a computer-based system to record and maintain information. The information concerned can be anything of significance to the organization for whose use it is intended. The database and the DBMS software together is a Database system.

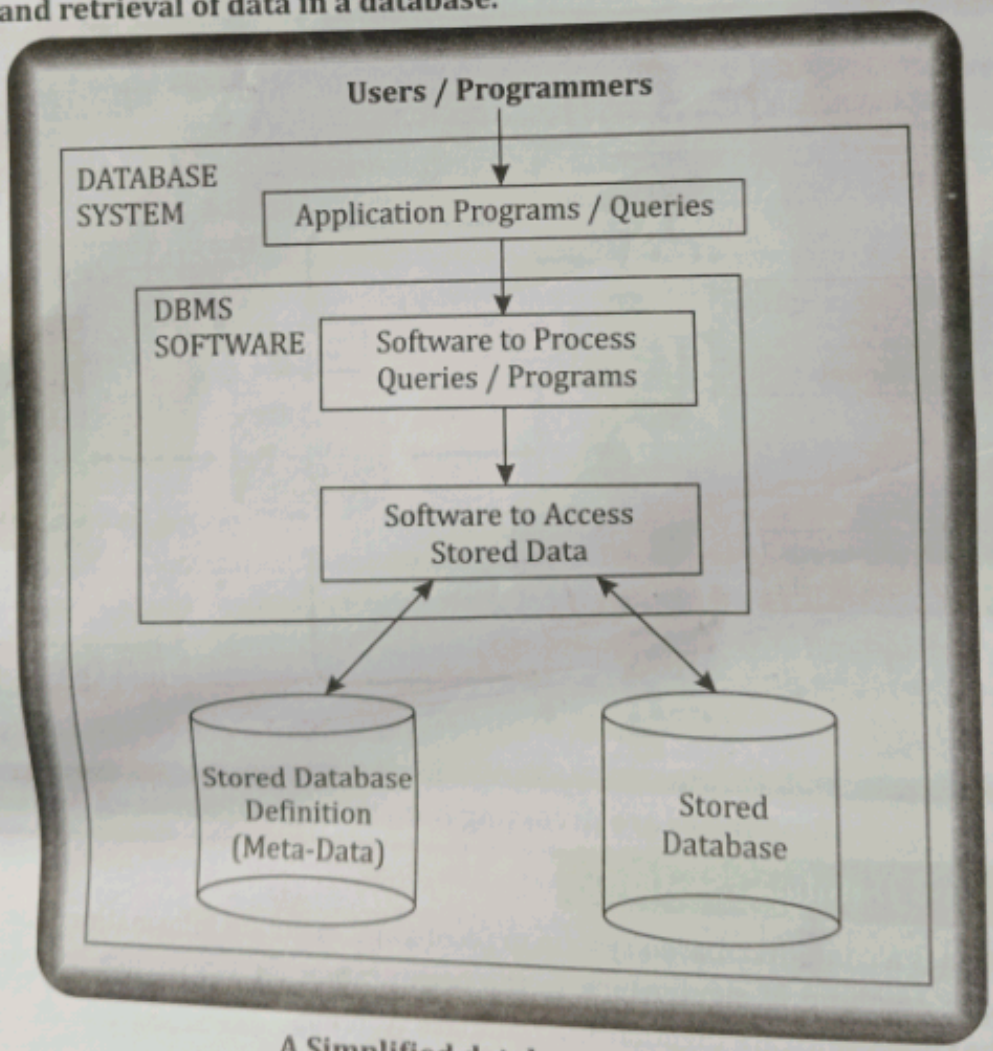
It consists of the following:

1. Users/Programmers
2. Applications Programs/Queries
3. Software to process Queries/Programs
4. Software to Access Stored data
5. DBMS catalog contains the stored database definition (Metadata)
6. The Physical stored database

A DBMS catalogue stores the *description* of the database. The description is called meta-data. This allows the DBMS software to work with different databases.

The DBMS interprets and processes users' requests to retrieve information from a database. A DBMS serves as an interface in several forms. They may be keyed directly from a terminal, or coded as high-level language programs to be submitted for interactive or batch processing. In most cases, a query request will have to penetrate several layers of software in the DBMS and operating system before the physical database can be accessed.

A DBMS is a set of software programs that controls the system organization, storage, management and retrieval of data in a database.



A Simplified database System

DBMS are categorized according to their data structures or types. The DBMS accepts requests for data from an application program and instructs the operating system to transfer the appropriate data. The queries and responses must be submitted and received according to a format that conforms to one or more applicable protocols. When a DBMS is used, information systems can be changed more easily as the organization's information requirements change. New categories of data can be added to the database without disruption to the existing system.

1.10 Functions of DBMS

The functions performed by a typical DBMS are the following:

- **Data Definition :** The DBMS provides functions to define the structure of the data in the application. These include defining and modifying the record structure, the type and size of fields and the various constraints/conditions to be satisfied by the data in each field.
- **Data Manipulation :** Once the data structure is defined, data needs to be inserted, modified or deleted. The functions which perform these operations are also part of the DBMS. These functions can handle planned and unplanned data manipulation needs. **Planned queries** are those which form part of the application. **Unplanned queries** are ad-hoc queries which are performed on a need basis.
- **Data Security & Integrity :** The DBMS contains functions which handle the security and integrity of data in the application. These can be easily invoked by the application and hence the application programmer need not code these functions in his/her programs. Data Recovery & Concurrency Recovery of data after a system failure and concurrent access of records by multiple users are also handled by the DBMS.
- **Data Dictionary Maintenance** : Maintaining the Data Dictionary which contains the data definition of the application is also one of the functions of a DBMS.
- **Performance**: Optimizing the performance of the queries is one of the important functions of a DBMS. Hence the DBMS has a set of programs forming the Query Optimizer which evaluates the different implementations of a query and chooses the best among them.
- Thus the DBMS provides an environment that is both convenient and efficient to use when there is a large volume of data and many transactions to be processed.

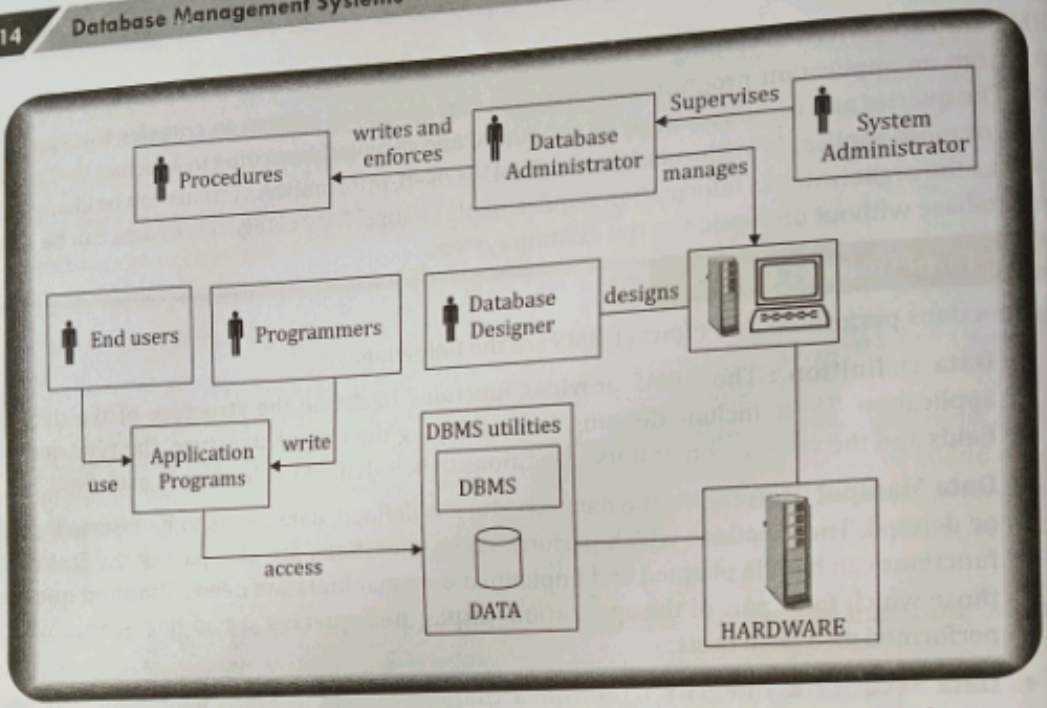
1.11 Database System Environment

The term database system refers to an organization of components that define and regulate the collection, storage, management, and use of data within a database environment. In a high level view the database system is composed of the following five major components:

1. Hardware
2. Software
3. People
4. Procedures
5. Data

1. **Hardware Components in a Database System Environment:**

   Hardware identifies all the system's physical devices. It includes computers, computer peripherals, network components etc.



2. **Software Components in a Database System Environment:** To make the database system work properly, three types of software are needed: operating system, DBMS software, and application programs.
   (a) **Operating System:** It manages all hardware components and allows other software to run on the computers. Examples of operating system software include Windows, Linux and etc.
   (b) **DBMS Software:** It manages the database within the database system. Some examples of DBMS software include Oracle, Access, MySql and etc.
   (c) **Application Programs:** These are used to access and manipulate data in the DBMS and to manage the computer environment in which data access and manipulation take place. Application programs are most commonly used to access data to generate reports. Most of the application programs provide GUI.
3. **People in a Database System Environment:** People component includes all users associated with the database system. On the basis of primary job function we can identify five types of users in a database system: System Administrators, Database Administrators, Data Models, System Analysts and Programmers and End Users.
   - System Administrators
   - Database Administrators
   - End Users
   - Data Modellers
   - System Analysts and Programmers

   (a) ***System Administrators*** oversees the database system's general operations.
   (b) ***Data Modellers*** (Architect) prepare the conceptual design from the requirement. ER model represent the conceptual design of an OLTP application.

(c) ***Database Administrator (DBA)*** physically implements the database according to the logical design. The DBA performs the physical implementation and maintenance of a database system.

(d) ***System Analysts and programmers*** design and implement the application programs. They create the input screens, reports, and procedures through which end users access and manipulate the database.

(e) ***End Users*** are the people who use the application. For example in case of a banking system, the employees, customer using ATM or online banking facility are end users.

4. **Procedures in a Database Environment:** Procedures are the instructions and rules that supervise the design and use of the database system. Procedures are a critical component of the system. Procedures play an important role in a company because they enforce the standards by which business is conducted in an organization.
5. **Data in the Database:** Data are the very important basic entity in a database. It is the collection of facts stored in the database. Data is the raw material from which information is generated; no database can exist without database.

1.12 People Associated with Database System (Database Users)

DBMS Users may be divided into:

1. Those who actually use and control the content called ***"Actors on the Scene"***.
2. Those who enable the database to be developed and the DBMS software to be designed and implemented called ***"Workers behind the Scene"***.

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1.12.1 Actors on the Scene

For a small personal database, one person typically defines, constructs, and manipulates the database, and there is no sharing, sharing of data is not required. However, many persons are involved in the design, use, and maintenance of a large database with hundreds of users. In this section we identify the people whose jobs involve in the day-to-day use of a large database.

1. **Database Administrator (DBA) :**

   In any organization where many persons use the same resources, there is a need for a chief administrator to oversee and manage these resources. In a database environment, the primary resource is the **database itself**, and the **secondary resource is the DBMS and related software**. Administering these resources is the responsibility of the **database administrator** (DBA). The DBA is responsible for authorizing access to the database, for coordinating and monitoring its use, and for acquiring software and hardware resources as needed. The DBA is accountable for problems such as breach of security or poor system response time.

2. **Database Designers :**

   Database designers are responsible for identifying the data to be stored in the database for choosing appropriate structures to represent and store this data. These tasks are mostly undertaken before the database is actually implemented and populated with data. It is the responsibility of database designers to communicate with all prospective database users in order to understand their requirements, and to come up with a design that meets these

requirements. In many cases, the designers are on the staff of the DBA and may be assigned other staff responsibilities after the database design is completed. Database designers typically interact with each potential group of users and develop views of the database that meet the data and processing requirements of these groups. Each view is then analyzed and *integrated* with the views of other user groups. The final database design must be capable of supporting the requirements of all user groups.

3. **End Users :**

End users are the people whose jobs require access to the database for querying, updating, and generating reports; the database primarily exists for their use. There are several categories of end users:

- ***Casual end users*** occasionally access the database, but they may need different information each time. They use a sophisticated database query language to specify their requests and are typically middle- or high-level managers or other occasional browsers.
- ***Naive or parametric end users*** make up a sizable portion of database end users. Their main job function revolves around constantly querying and updating the database, using standard types of queries and updates-called canned transactions-that have been carefully programmed and tested.

The tasks that such users perform are varied

1. Bank tellers check account balances and post withdrawals and deposits.
2. Reservations clerks fur airlines, hotels, and car rental companies check availability for given request and make reservations.
3. Clerks at receiving stations for courier mail enter package identifications via bar codes and descriptive information through buttons to update a central database of received and in-transit packages.

- ***Sophisticated end users*** include engineers, scientists, business analysts, and others who thoroughly familiarize themselves with the facilities of the DBMS so as to implement their applications to meet their complex requirements.
- ***Stand-alone users*** maintain personal databases by using ready-made program packages that provide easy-to-use menu-based or graphics-based interfaces. An example is the user of a tax package that stores a variety of personal financial data for tax purposes.

A typical DBMS provides multiple facilities to access a database. ***Naive end users*** need to learn very little about the facilities provided by the DBMS; they have to understand only the user interfaces of the standard transactions designed and implemented for their use. ***Casual users*** learn only a few facilities that they may use repeatedly. ***Sophisticated users*** try to learn most of the DBMS facilities in order to achieve their complex requirements. Stand-alone users typically become very proficient in using a specific software package.

4. **System Analysts and Application Programmers (Software Engineers)**

System analysts determine the requirements of end users, especially naive and parametric end users, and develop specifications for canned transactions that meet these requirements. Application programmers implement these specifications as programs; then they test, debug,

document, and maintain these canned transactions. Such analysts and programmers-commonly referred to as software engineers-should be familiar with the full range of capabilities provided by the DBMS to accomplish their tasks.

1.12.2 Workers Behind the Scene

In addition to those who design, use, and administer a database, others are associated with the design, development, and operation of the DBMS *software and system environment*. These persons are typically not interested in the database itself. We call them the "workers behind the scene," and they include the following categories.

1. **DBMS System Designers and Developers** are persons who design and implement the DBMS modules and interfaces as a software package. A DBMS is a very complex software system that consists of many components, or modules, including modules for implementing the catalog, processing query language, processing the interface, accessing and buffering data, controlling concurrency, and handling data recovery and security. The DBMS must interface with other system software, such as the operating system and compilers for various programming languages.
2. **Tool Developers** include persons who design and implement tools-the software packages that facilitate database system design and use and that help improve performance. Tools are optional packages that are often purchased separately. They include packages for database design, performance monitoring, natural language or graphical interfaces, prototyping, simulation, and test data generation. In many cases, independent software vendors develop and market these tools.
3. **Operators and Maintenance Personnel** are the system administration personnel who are responsible for the actual running and maintenance of the hardware and software environment for the database system. Although these categories of workers behind the scene are instrumental in making the database system available to end users, they typically do not use the database for their own purposes.

1.13 Database Administrator (DBA)

Database Administrator (DBA) Coordinates all the activities of the database system and has a good understanding of the enterprise's information resources and needs.

Definition: Database Administration

Database administration is more of an operational or technical level function responsible for physical database design, security enforcement, and database performance. Tasks include maintaining the data dictionary, monitoring performance, and enforcing organizational standards and security.

? What is the role of Database Administrator ?

The person who is responsible for controlling and coordinating the DBMS system is called Database System Administrator (DBA).

1.13.1 Roles and Responsibilities of DBA

The job of the DBA seems to be everything that everyone else either doesn't want to do, or doesn't have the ability to do. DBAs get the enviable task of figuring out all of the things no one else can figure out. More seriously though, here is a list of typical DBA responsibilities:

1. Installation, configuration and upgrading of DBMS software and related products
2. Evaluate DBMS features and related products
3. Establish and maintain sound backup and recovery policies and procedures
4. Take care of the Database design and implementation
5. Implement and maintain database security (create and maintain users and roles, assign privileges)
6. Perform database tuning and performance monitoring
7. Perform application tuning and performance monitoring
8. Setup and maintain documentation and standards
9. Plan growth and changes (capacity planning)
10. Work as part of a team and provide 24 x 7 support when required
11. Perform general technical trouble shooting and give consultation to development teams
12. Interface with DBMS Vendors for technical support.
13. Patch Management and Version Control

1.13.2 Functions of Database Administrator

1. **Selection of hardware and software**
   - Keep up with current technological trends
   - Predict future changes
   - Emphasis on established off the shelf products
2. **Managing data security and privacy**
   - Protection of data against accidental or intentional loss, destruction, or misuse
   - Fire walls
   - Establishment of user privileges
   - Complicated by use of distributed systems such as internet access and client/ server technology.

1.14 Database Applications

Database systems are widely used in almost all fields. Due to their numerous advantages, database systems have been successfully used thy the following applications:

- **Railway Reservation System :** The database is required to keep record of ticket booking, train departure, and arrival status.

- **Library Management System :** There are thousands of books in the library so it is very difficult to keep record of all the books in a copy or register. So DBMS is used to maintain all the information related to book issuing dates, name of the book, author, and availability of the book.
- **Banking :** Financial institution (such as banks) use data base for managing the records of customers, accounts, loans, and other transactions. The databses are used for recording credit card purchase for generating monthly bills,
- **Education Sector - Schools, Colleges and Universities :** The information about students' registration numbers, courses opted, fees structure, date of birth, address etc. all of the data is fed into DBMS and is retrieved with the help of the Database Admin.
- **Telecommunications :** Any telecommunication company cannot even think about its business without DBMS. DBMS is a must for these companies to store the customer details, call details and monthly post-paid bills.
- **Online Shopping :** These days, web-based shopping has become a major pattern. Nobody needs to visit the shop. Everybody needs to shop through web based shopping sites, (for example, Amazon, Flipkart, Snapdeal) from home. These companies sell a variety of products and each product has a price, description, and other characteristics. Multiple orders can be booked via a single user. When an order is shipped and delivered all the necessary details like Invoice, Shipping details, etc. are automatically stored in the database.
- **Human Resource Management :** The big companies have a lot of employees and representatives working under them. All the necessary information i.e. employees assessment, tax records, salary slip, etc are maintained and stored in DBMS.
- **Manufacturing :** Manufacturing companies make products and sell them on the daily basis. Databases are used to keep records of all the details about the products like quantity, bills, purchase, supply chain management.
- **Internet :** The World Wide Web maintains enormous amount of textual and multimedia data in database.
- **Online Trading :** Databases are used in online trading for storing information about the sales and purchases of stocks and bonds.
- **Military :** The military keeps records of millions of soldiers and it has millions of files that should be kept secured and safe. Database systems provide a big security assurance to the military information so it is widely used in militaries. One can easily search for all the information about anyone within seconds with the help of DBMS.

1.15 Data Model

Data Model is a method used to define and analyze data requirements needed to support the business processes of an organization and by defining the data structures and the relationships between data elements.

A data model is a picture or description which depicts how data is to be arranged to serve a specific purpose. The model provides the necessary means to achieve the abstraction.

Data Abstraction hides details of data storage that are not needed by most database users and applications.

A model is a representation of reality, **'real world'** objects , events and associations. It is an abstraction that concentrates on the essential and inherent aspects of an organization. A data model represents the organization itself. It should provide the basic concepts and notations that will allow database designers and end users to communicate their understanding of the organizational data unambiguously and accurately.

The main purpose of a data model is to represent data and to make the data understandable.



Definition: Data Model

Data Model can be defined as an integrated collection of concepts for describing and manipulating data, relationships between data, and constraints on the data in an organization.

A data model comprises of three components:

- **Structural Part :** consisting of a set of rules according to which databases can be constructed.
- **Manipulative Part:** defining the types of operation that are allowed on the data (this includes the operations that are used for updating or retrieving data from the database and for changing the structure of the database).
- **Set of Integrity Rules :** Which ensures that the data is accurate.

Data Modeling techniques are used

- To manage data as a resource (migrate, merge, data quality)
- For designing computer databases.
- To make changes into the data model, that will automatically induce changes in the database and programs.

1.15.1 Categories of Data Models

Data models can be categorized in multiple ways

1. **High Level/Conceptual Data Models** – provide concepts close to the way users perceive the data, use concepts such as entities, attributes, relationships. Object-based models Such as ER model, OO model.

   Conceptual data models use concepts such as Entities, Attributes and Relationships:

   - **Entity** – represents a real world object or concept.
   - **Attribute** - represents property of interest that describes an entity, such as name or salary.
   - **Relationships** – among two or more entities, represents an association among two or

2. **Low Level/Physical Data Models** – provide concepts that describe the details of how data is stored in the computer. These concepts are generally meant for the specialist, and not the end user. ***Physical data models*** captures aspects of database system implementation and describe how data is stored in files by representing record formats (fixed/variable length), record orderings and access paths (key indexing) etc.

3. **Representational/Implementation Data Models** – provide concepts that may be understood by the end user but not far removed from the way data is organized. ***Representational data models*** are used most frequently in commercial DBMSs. They include ***Relational data models***, and Legacy models such as ***Network and Hierarchical models.***

1.15.2 Different Types of Data Models

There are five models of DBMS, which are distinguished based on how they represent the data contained:

1. Hierarchical Data Model
2. Network Data Model
3. Relational Data Model
4. Object-Oriented Data Model
5. Object -Relational Data Model

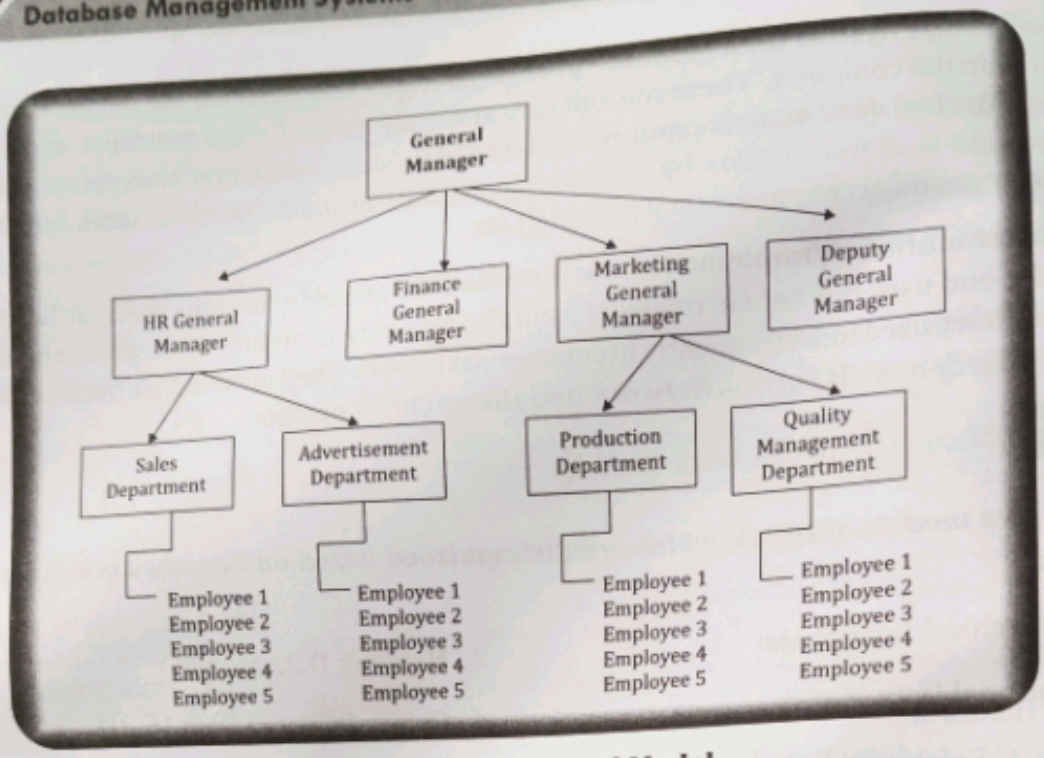
1.15.2.1 Hierarchical Data Model

Hierarchical Database model is one of the oldest database models, dating from late 1950s. One of the first hierarchical databases Information Management System (IMS) was developed jointly by North American Rockwell Company and IBM.

The hierarchical data model organizes data in a tree structure. There is a hierarchy of parent and child data segments. This model is like a structure of a tree with the records forming the nodes and fields forming the branches of the tree. This structure implies that a record can have repeating information, generally in the child data segments. Data in a series of records, which have a set of field values attached to it. It collects all the instances of a specific record together as a record type. These record types are the equivalent of tables in the relational model, and with the individual records being the equivalent of rows. To create links between these record types, the hierarchical model uses Parent Child Relationships. These are a 1:N mapping between record types. The hierarchical model organizes data elements as tabular rows, one for each instance of entity.

In the hierarchical data model, records are linked with other superior records on which they are dependent and also on the records, which are dependent on them. A tree structure may establish one-to-many relationship.

The different elements (example: records) present in the hierarchical tree structure have Parent-Child relationship. A Parent element can have many children elements but a Child element cannot have many parent elements. That is, hierarchical model cannot represent many to many relationships among records.



Example of Hierarchical Model

Example: **Consider a company's organizational structure. At the top we have a General Manager (GM).**

Under him we have several Deputy General Managers (DGMs). Each DGM looks after a couple of departments and each department will have a manager and many employees. When represented in hierarchical model, there will be separate rows for representing the GM, each DGM, each department, each manager and each employee. The row position implies a relationship to other rows. A given employee belongs to the department that is closest above it in the list and the department belongs to the manager that is immediately above it in the list and so on as shown in the above figure.

Definition: *Hierarchical Data Model*

The hierarchical data model organizes data in a tree structure. There is a hierarchy of parent and child data segments.

Advantages of Hierarchical Model

- **Simplicity :** Data naturally have hierarchical relationship in most of the practical situations. Therefore, it is easier to view data arranged in tree structure manner.
- **Security :** These database system can enforce varying degree of security feature unlike flat file system.
- **Database Integrity :** Because of its inherent parent-child structure, database integrity is highly promoted in these systems.
- **Efficiency:** The hierarchical database model is a very efficient one when the database contains a large number of 1 : N relationships (one-to-many relationships) and when the users require large number of transactions, using data whose relationships are fi

Disadvantages of Hierarchical Model

- **Implementation Complexity:** Although it is simple and easy to design, it is quite complex to implement.
- **Database Management Problem:** If we make any changes in the database structure, then we need to make changes in the entire application program that access the database.
- **Lack of Structural Independence:** There is a lack of structural independence because when we change the structure then it becomes compulsory to change the application too.
- **Operational Anomalies:** Hierarchical model suffers from the insert, delete and update anomalies, also retrieval operation is difficult.

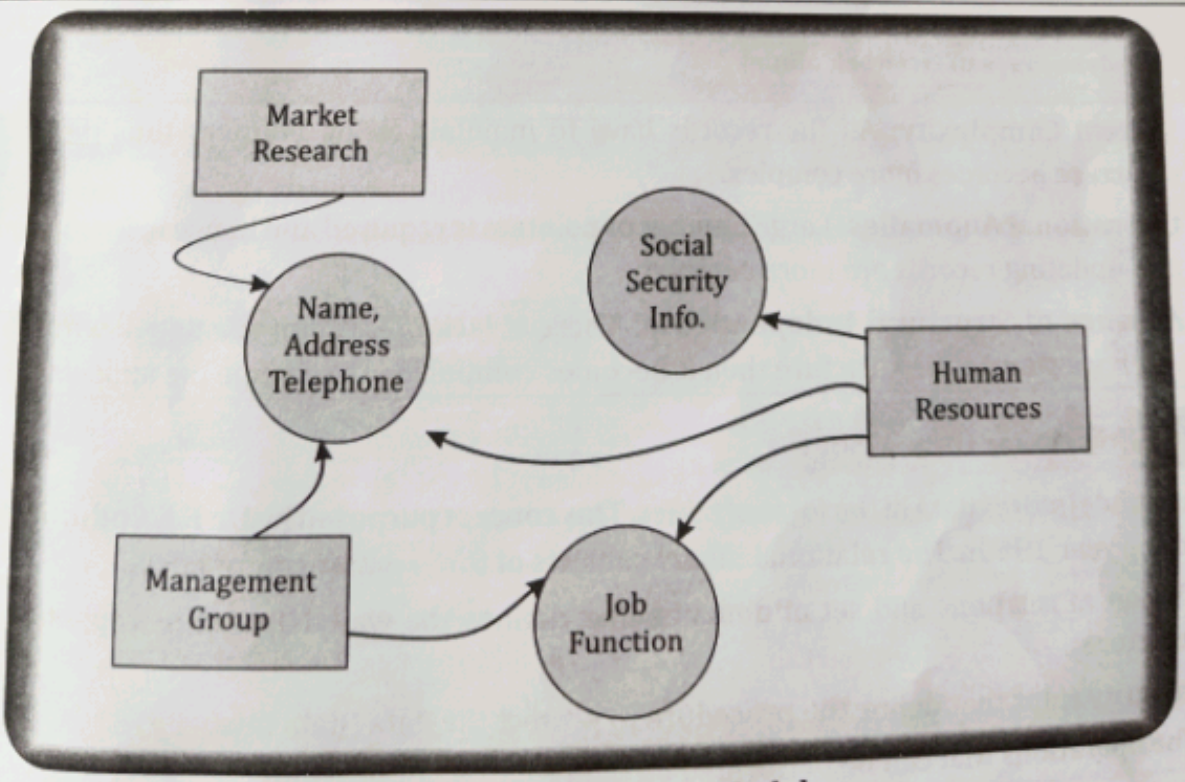
1.15.2.2 Network Data Model

The Network model replaces the hierarchical tree with a graph thus allowing more general connections among the nodes. The main difference of the network model from the hierarchical model, is its ability to handle many to many relations. In other words, it allows a record to have more than one parent.

Network model is a collection data in which records are physically linked through linked lists. A DBMS is said to be a Network DBMS if the relationships among data in the database are of type many-to-many. The relationship among many-to-many appears in the form of a network.

Thus the structure of a network database is extremely complicated because of the many-to-many relationships in which one record can be used as a key of the entire database. A network database is structured in the form of a graph that is also a data structure.

Example : **Consider an example of HR Manager where Human Resources is owner to all 3 records. Market Research and Management Group are owners to only certain members.**



Example of Network Model

Suppose an employee works for two departments. The strict hierarchical arrangement is not possible in this case and the tree becomes a more generalized graph – a network.

The network model was evolved to specifically handle non-hierarchical relationships. A network structure thus allows 1:1 (one:one), 1:N (one:many), N:N (many:many) relationships among entities.

In network database terminology, a relationship is a set. Each set is made up of at least two types of records: an owner record (equivalent to parent in the hierarchical model) and a member record (similar to the child record in the hierarchical model).

Definition: Network Data Model

The network model organizes data using two fundamental constructs called records and sets. Records contain fields, Sets define one-to-many relationships between records.

Advantages of Network Model

- **Conceptual Simplicity:** Just like hierarchical model it also simple and easy to implement.
- **Capability to Handle more Relationship Types:** The network model can handle one to one(1:1) and many to many (N: N) relationship.
- **Ease to Access Data:** The data access is easier than the hierarchical model.
- **Data Integrity:** Since it is based on the parent child relationship, there is always a link between the parent segment and the child segment under it.
- **Data Independence:** The network model is better than hierarchical model in case of data independence.

Disadvantages of Network Model

- **System Complexity:** All the records have to maintain using pointers thus the database structure becomes more complex.
- **Operational Anomalies:** Large number of pointers is required and hence insertion, deletion and updating records are more complex.
- **Absence of Structural Independence:** There is lack of structural independence because when we change the structure then it becomes compulsory to change the application too.

1.15.2.3 Relational Data Model

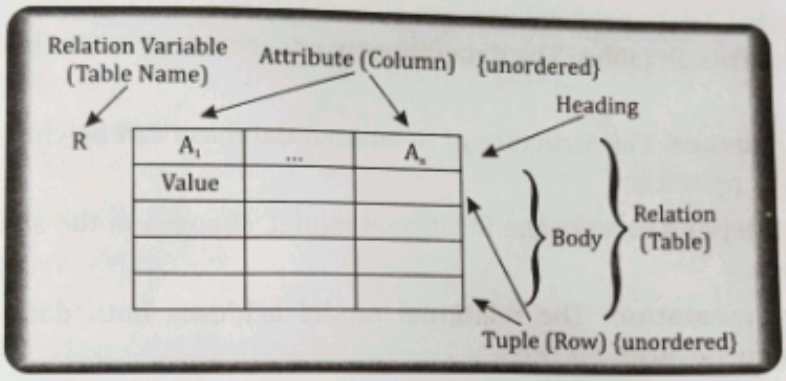
Relational model stores data in the form of tables. This concept purposed by **Dr. E.F. Codd,** a researcher of IBM in the year 1960s. The relational model consists of three major components:

1. The set of relations and set of domains that defines the way data can be represented (data structure).
2. Integrity rules that define the procedure to protect the data (data integrity).
3. The operations that can be performed on data (data manipulation).

Definition: Relational Model

A Relational model is defined as a database that allows to group its data items into one or more independent tables that can be related to one another by using common fields related to each table.

The Relational Model *(RDBMS, Relational database management system):* The data is stored in two-dimensional tables (rows and columns). The data is manipulated based on the relational theory of mathematics.



Example of Relational Model

Characteristics of Relational Database

- The whole data is conceptually represented as an orderly arrangement of data into rows and columns, called a **relation** or **table**.
- All values are scalar. That is, at any given row/column position in the relation there is one and only one value.
- All operations are performed on an entire relation and result is an entire relation, a concept known as closure.

Example : **The Relational Model for database management is a database model based on first-order predicate logic.**

Student

Reg_No	Std_Name	Course	Email_id
5001	Kavitha	BCA	kavitha@gmail.com
5002	Shwetha	BBM	shwetha@gmail.com
5003	Shruthi	BCA	shruthi@gmail.com

Marks

Reg_No	Course	Result
5001	BCA	Distinction
5002	BBM	First class
5003	BCA	First class

Performance Analysis

Reg_No	Std_Name	Course	Result
5001	Kavitha	BCA	Distinction
5002	Shwetha	BBM	First class
5003	Shruthi	BCA	First class

The related records are linked together with a "key" in relational model.

Advantages of Relational Model (Relational Databases)

- **Simplicity:** We have seen that both the hierarchical and network models are conceptually simple, but relational model is simpler than both of those two.
- **Ease of Use:** Users can easily access/retrieve the required information from the database. The Relational model in DBMS is easy as tables consisting of rows and columns are quite natural and simple to understand
- **Data Integrity :** Relational databases are also widely used for data integrity as they provide consistency across all tables. The data integrity ensures the features like accuracy and ease of use.
- **Data independence:** The Structure of Relational database can be changed without having to change any application.
- **Structural Independence:** In the Relational model, changes in the structure do not affect the data access.
- **Design Implementation:** The relational model achieves both data independence and structural independence.
- **Security :** Data is secure as Relational Database Management System allows only authorized users to directly access the data. No unauthorized user can access the information.
- **Collaboration :** Multiple users can access the database to retrieve information at the same time and even if data is being updated.

Disadvantages of Relational Model

- **Maintenance Problem :** The maintenance of the relational database becomes difficult over time due to the increase in the data. Developers and programmers have to spend a lot of time in maintaining the database.
- **Cost :** The relational database system is costly to set up and maintain. The initial cost of the software alone can be quite high for smaller businesses.
- **Physical Storage :** A relational database is comprised of rows and columns, which requires a lot of physical memory because each operation performed depends on separate storage. The requirements of physical memory may increase along with the increase of data.
- **Lack of Scalability :** While using the relational database over multiple servers, its structure changes and becomes difficult to handle, especially when the quantity of the data is large. Due to this, the data is not scalable on different physical storage servers. Ultimately, its performance is affected i.e. lack of availability of data and load time etc.
- **Complexity in Structure :** Relational databases can only store data in tabular form which makes it difficult to represent complex relationships between objects. This is an issue because many applications require more than one table to store all the necessary data required by their application logic.

1.15.2.4 Object-Oriented Data Model

To represent the complex real world problems there was a need for a data model that is closely related to real world. Object Oriented Data Model represents the real world problems easily.

In Object-Oriented Programming, everything is an object, and many objects are quite complex, having different properties and methods. An object-oriented database management system works in concert with an object-oriented programming language to facilitate the storage and retrieval of object-oriented data.

In Object Oriented Data Model, data and their relationships are contained in a single structure which is referred as object. In this, real world problems are represented as objects with different attributes. All objects have multiple relationships between them.

Object-oriented databases directly deal with data as complete objects. All the information comes in one instantly available object package instead of multiple tables.

A object-oriented data model consists of the following basic object-oriented concepts:

1. **Object:** The real world entities and situations are represented as objects in the Object oriented database model.
2. **Attributes and Methods:** Every object has certain characteristics. These are represented using Attributes. The behaviour of the objects is represented using Methods.
3. **Class:** Similar attributes and methods are grouped together using a class. An object can be called as an instance of the class..
4. **Inheritance:** A new class can be derived from the original class. The derived class contains attributes and methods of the original class as well as its own.

These characteristics make object databases suitable for projects with complex data which require an object-oriented approach to programming. An object-oriented management system provides supported functionality catered to object-oriented programming where complex objects are central. This approach unifies attributes and behaviors of data into one entity.



Definition: Object Oriented Data Model

Object-oriented databases (OODB) are databases that represent data in the form of objects and classes. In object-oriented terminology, an object is a real-world entity, and a class is a collection of objects. Object-oriented databases follow the fundamental principles of object-oriented programming (OOP). The combination of relational model features (concurrency, transaction, and recovery) with object-oriented principles results in an object-oriented database model.



Advantages of Object Oriented Data Model

- Complex data sets can be saved and retrieved quickly and easily.
- It works well with object-oriented programming languages.
- Unlike traditional databases (such as hierarchical, network or relational), the object oriented database are capable of storing different types of data, for example, pictures, voice video, including text, numbers and so on.

- It allow new data types to be built from existing types. The ability to factor out common properties of several classes and form them into a super-class that can be shared with sub-classes can greatly reduce redundancy within system is regarded as one of the main advantages of object orientation.
- There have been a number of benchmarks that have suggested that Object Orineted Database Systems provide significant performance improvements over relational DBMSs.

Disadvantages of Object Oriented Data Model

- There is no universally agreed data model for an OODBMS, and most models lack a theoretical foundation.
- In comparison to relational database systems the use of object oriented database is still relatively limited. This means that we do not yet have the level of experience that we have with traditional systems.
- There is no standard object-oriented query language.

1.15.2.5 Object Relational Data Model

An Object relational model is a combination of a Object oriented database model and a Relational database model. So, it supports objects, classes, inheritance etc. just like Object Oriented models and has support for data types, tabular structures etc. like Relational data model.

One of the major goals of Object relational data model is to close the gap between relational databases and the object oriented practises frequently used in many programming languages such as C++, C#, Java etc.

Object Relational Data model can be thought of as an attempt to extend relational database systems with the functionality necessary to support a broader class of application domains, provide a bridge between the relational and object-oriented paradigms. This approach attempts to get the best of both.

Definition: Object Relational Data Model

An object-relational database (ORD) is a database management system (DBMS) that's composed of both a relational database (RDBMS) and an object-oriented database (OODBMS). ORD supports the basic components of any object-oriented database model in its schemas and the query language used, such as objects, classes and inheritance.

An object-relational database may also be known as an object relational database management systems (ORDBMS).

Advantages of Object Relational Data Model

- **Inheritance :** The Object Relational data model allows its users to inherit objects, tables etc. so that they can extend their functionality. Inherited objects contains new attributes as well as the attributes that were inherited.

- **Complex Data Types :** Complex data types can be formed using existing data types. This is useful in Object relational data model as complex data types allow better manipulation of the data.
- **Extensibility :** The functionality of the system can be extended in Object relational data model. This can be achieved using complex data types as well as advanced concepts of object oriented model such as inheritance.

Disadvantages of Object Relational Data Model

- The ORDBMSs approach has the obvious disadvantage of complexity and associated increased costs.
- There are proponents of the relational approach that believe the essential simplicity and purity of the relational model are lost with these types of extension.

1.16 Schemas and Instances

1.16.1 Schema

A database is a place to store information. It can store the simplest data, such as a list of people as well as the most complex data. The database stores the information in a well-structured format. In any data model, it is important to distinguish between the description of the database and the database itself. Schema is the overall description of the database. The basic structure of how the data will be stored in the database is called schema.

Definition: Database Schema

A database schema is the logical representation of a database, which shows how the data is stored logically in the the database. It contains list of attributes and instruction that informs the database engine that how the data is organized and how the elements are related to each other.

The schema does not physically contain the data itself; instead, it gives information about the shape of data and how it can be related to other tables or models.

1.16.2 Database Instance

- Current contents of the database (actual data): Database state
- May change frequently

The data in the database at a particular moment in time is called a database state or snapshot. It is also called the *current* set of occurrences (or) ***instances*** in the database. In a given database state, each schema construct has its own *current set* of instances.

Instances are the collection of information stored at a particular moment. The instances can be changed by certain operations as like addition, deletion of data.

Example, the Student construct will contain the set of individual student entities (records) as its instances. Suppose the table has 50 records so the instance of the database has 50 records for now and tomorrow we are going to add another 30 records so tomorrow the instance have total 100 records. This is called an instance.

Definition: Database Instance (or) Database State

The data in the database at a particular moment in time is called the Database state (or) Snapshot (or) instance.

- When a database is defined, the schema is specified to the DBMS. The database state at this point is in the ***empty state***, with no data.
- The ***initial state*** of the database is when the database is first populated or loaded with the initial data. Every time data is added/removed/updated, there is a new database instance.
- The DBMS is responsible for ensuring every state is a ***valid state***, a state that satisfies the structure and constraints specified in the schema.
- Both instances and schemas are related to and impact each other through the DBMS. DBMS ensures that every database instance complies with the constraints imposed by the database designers in the database schema.
- The DBMS stores the descriptions of the schema constructs and constraints, called the ***meta-data***, in the DBMS catalogue.
- The schema is called the ***Intension of the schema*** and the database state an ***Extension of the schema.***

1.16.3 Database Schema Vs Database State

Database Schema	Database Instance
It is the overall description of the database.	It is the collection of information stored in a database at a particular moment.
Schema is same for whole database.	Data in instances can be changed using addition, deletion, updation.
Does not change frequently.	Changes frequently.
Defines the basic structure of the database i.e how the data will be stored in the database.	It is the set of Information stored at a particular time.

1.16.4 Schema Diagram

A schema diagram is a diagram which contains entities and the attributes that will define that schema. A schema diagram only shows us the database design. It does not show the actual data of the database. Schema can be a single table or it can have more than one table which is related. The schema represents the relationship between these tables. Each object in the schema is called a **Schema Construct**.

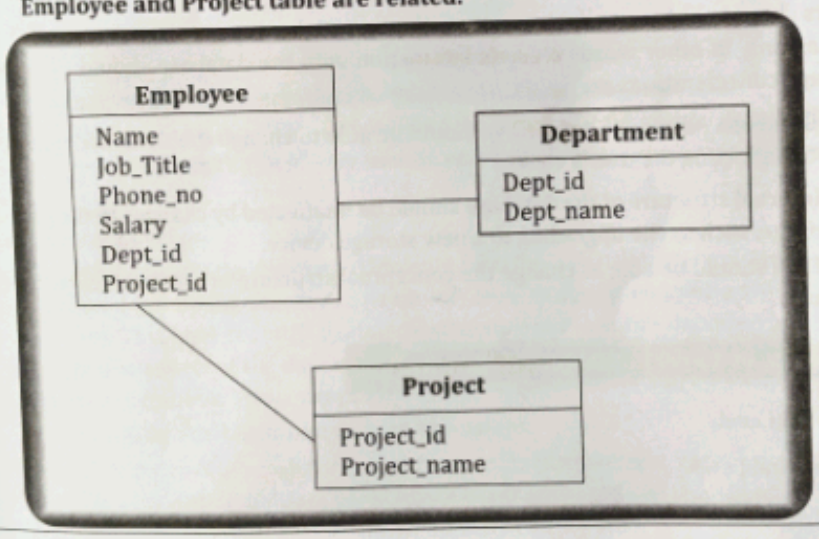
Example: Imagine a case where we want to store facts about employees in a company. Such facts could include their name, address, date of birth, and salary. In a database all the information on all employees would be held in a single storage "container", called a Table. This table is a tabular object like a spreadsheet page, with different employees as the rows, and the facts (e.g. their names) as columns... Let's call this table EMP and it could look something like:

Name	Address	Date of Birth	Salary

From this information the **schema** would define that **EMP** has four components, "NAME","ADDRESS","DOB","SALARY".

1. **NAME** is a string, and needs to hold at least 12 characters.
2. **ADDRESS** is a string, and needs to hold at least 12 characters.
3. **DOB** is a date. It should be in dd/mm/yyyy format.
4. **SALARY** is a number. It must be greater than zero.

Example: **Let us suppose we have three tables Employee, Department and Project. So, we can represent the schema of these three tables using the schema diagram as follows. In this schema diagram, Employee and Department are related and the Employee and Project table are related.**



1.17 DBMS Architecture

Database Management Systems are very complex, sophisticated software applications that provide reliable management of large amounts of data. To better understand general database concepts and the structure and capabilities of a DBMS, it is useful to examine the architecture of a typical database management system.

There are two different ways to look at the architecture of a DBMS: ***The logical DBMS Architecture (Three schema Architecture)*** and ***The physical DBMS architecture***. The logical architecture deals with the way data is stored and presented to users, while the physical architecture is concerned with the software components that make up a DBMS.

1.17.1 Logical DBMS Architecture (Three Level or Schema DBMS Architecture)

The logical architecture describes how data in the database is perceived by users. It is not concerned with how the data is handled and processed by the DBMS, but only with how it looks. Users are shielded from the way data is stored on the underlying file system, and can manipulate the data without worrying about where it is located or how it is actually stored. This results in the database having different levels of abstraction.

Objective of Three Level Architecture

The objective of the three level architecture is to separate each user's view of the database from the way the database is physically represented. There are several reasons why this separation is desirable:

1. Each user should be able to access the same data, but have a different customized view of the data. Each user should be able to change the way he or she views the data, and this change should not affect other users.
2. Users should not have to deal directly with physical database storage details, such as indexing or hashing. In other words a user's interaction with the database should be independent of storage considerations.
3. The Database Administrator (DBA) should be able to change the database storage structures without affecting the user's views.
4. The internal structure of the database should be unaffected by changes to the physical aspects of storage, such as the upgrading to a new storage device.
5. The DBA should be able to change the conceptual structure of the database without affecting all users.

Three Levels or Layers of DBMS Architecture

(a) External Level

(b) Conceptual Level

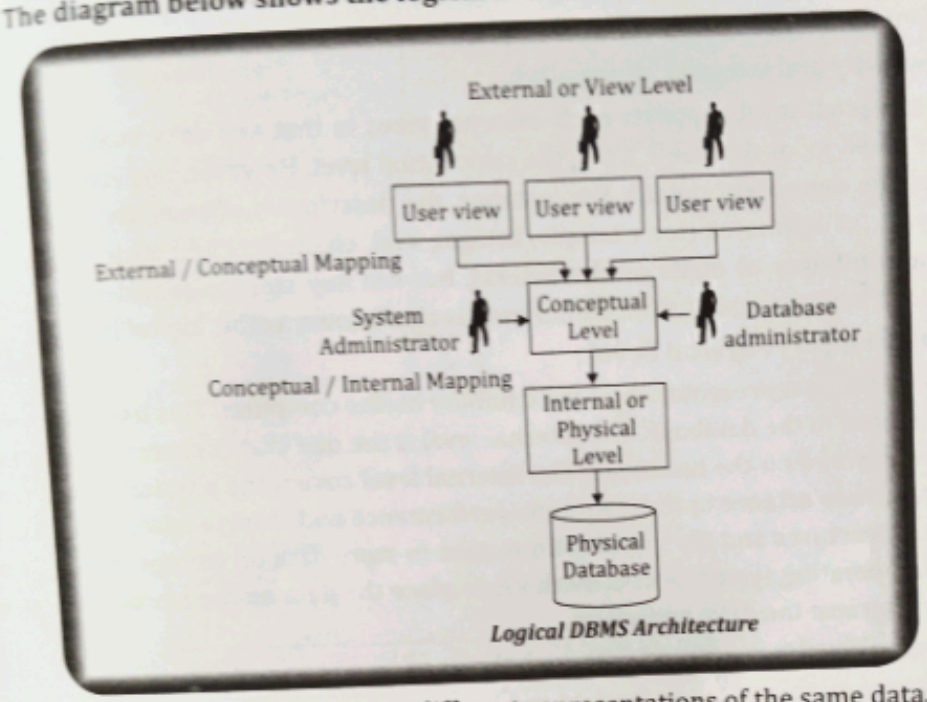
(c) Internal Level

1. The External (or) View Level :

It is the users' view of the database. This level describes that part of the database that is relevant to each user. External level is the one which is closest to the end users. This level deals with the way in which individual users view the data. Individual users are given different views according to the user's requirement.

A view involves only those portions of a database which are of concern to a user. Therefore same database can have different views for different users. The external view insulates users from the details of the internal and conceptual levels. External level is also known as the view

The diagram below shows the logical architecture for a typical DBMS:



Logical DBMS Architecture

In addition different views may have different representations of the same data. For example, one user may view dates in the form (day, month, year), while another may view dates as (year, month, day).

2. **The Conceptual Level :**

It is the community view of the database. This level describes what data is stored in the database and the relationships among the data. The middle level in the three level architecture is the conceptual level. This level contains the logical structure of the entire database as seen by the DBA. It is a complete view of the data requirements of the organization that is independent of any storage considerations. The conceptual level represents:

- All entities, their attributes, and their relationships;

An Entity is an object whose information is stored in the database. For example, in student is an entity in student database. An attribute is a characteristic of interest about an entity. Roll No, Name, Class, Address etc. are attributes of entity student.

Field Name ↓

Roll No.	Name	Class	Address
101	Naven	BCA	Jayanagar
102	Nandas	BSc	Hanumanthanagar
103	Vinaya	BCom	I.P.N
104	Nandish	BBA	N R Colony

↑ Field of Column or attribute

- The constraints on the data;
- Semantic information about the data;
- Security and integrity information.

The conceptual level supports each external view, in that any data available to a user must be contained in, or derivable from, the conceptual level. However, this level must not contain any storage dependent details. For instance, the description of an entity should contain only data types of attributes (for example, integer, real, character) and their length (such as the maximum number of digits or characters), but not any storage considerations, such as the number of bytes occupied. Conceptual level is also known as the, logical level.

3. **The Internal (or) Physical Level :**

It is the physical representation of the database on the computer. This level describes how the data is stored in the database. The internal level is the one that concerns the way the data are physically stored on the hardware. The internal level covers the physical\ implementation of the database to achieve optimal runtime performance and storage space utilization. It covers the data structures and file organizations used to store data on storage devices. It interfaces with the operating system access methods to place the data on the storage devices, build the indexes, retrieve the data, and so· on.

The internal level is concerned with such things as:

- Storage space allocation for data and indexes;
- Record descriptions for storage (with stored sizes for data items);
- Record placement;
- Data compression and data encryption techniques.

There will be only one conceptual view, consisting of the abstract representation of the database in it's entirely. Similarly there will be only one internal or physical view, representing the total database, as it is physically stored.

1.17.2 Mapping

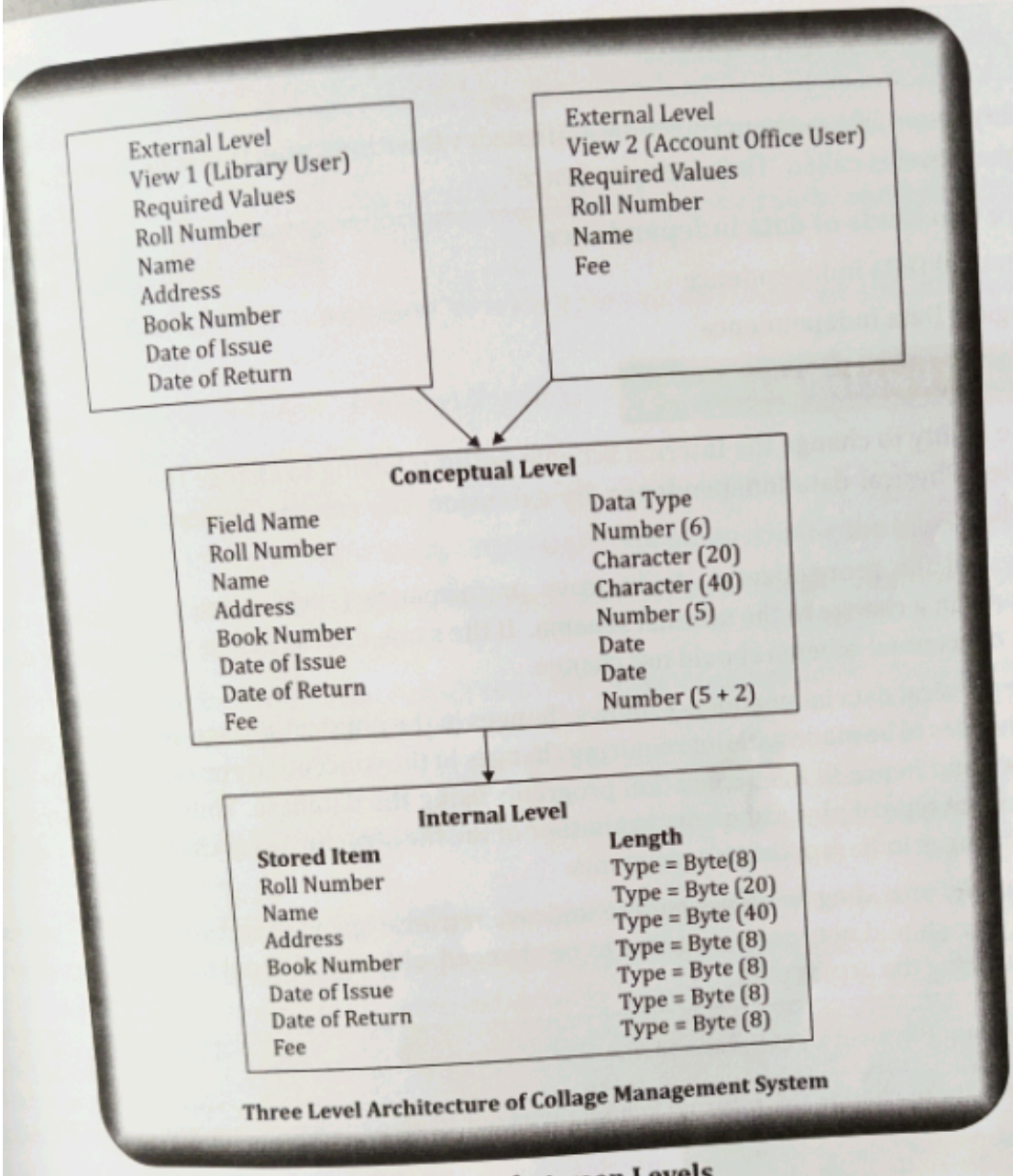
DBMS must transform a request specified on an external schema into a request against the conceptual schema and then into the internal schema.

Definition: Mapping

The processes of transforming requests and results between different levels of schema is called mappings

The DBMS is responsible for mapping between these three types of schema. Two mappings are required in a database system with three different views.

External/Conceptual Mapping: Each external schema is related to the conceptual schema by the external/conceptual mapping. A mapping between the external and conceptual views gives the correspondence among the records and the relationships of the external and conceptual views the external view is an abstraction of the conceptual view, which in its turn is an abstraction of the internal view. It describes the contents of the database as perceived by the user or application program of that view. The user of the external view sees and manipulates a record corresponding to the external view



Three Level Architecture of Collage Management System

Mappings between Levels

There is a mapping from a particular logical record in the external view to one (or more) conceptual record(s) in the conceptual view.

Conceptual/Internal Mapping: Conceptual schema is related to the internal schema by the conceptual/internal mapping. This enables the DBMS to find the actual record or combination of records in physical storage that constitute a logical record in conceptual schema. Mapping between the conceptual and the internal levels specifies the method of deriving the conceptual record from the physical database.

1.18 Data Independence

Data independence is one of the main advantages of DBMS. The three-schema architecture provides the concept of data independence, which means that upper-levels are unaffected by changes to lower levels. The three schemas architecture makes it easier to achieve true data independence.

Definition: Data Independence

The ability to modify a schema definition in one level without affecting schema definition in the next higher level is called "Data Independence".

There are two kinds of data independence

1. Physical Data independence
2. Logical Data independence

1. Physical Data Independence

- The ability to change the internal schema without having to change the conceptual schema is called Physical data independence. By extension, the external schema should not change as well.
- Physical file reorganization to improve performance (such as creating access structures) results in a change to the internal schema. If the same data as before remains in the database, the conceptual schema should not change.
- The physical data independence allows changes in the physical storage devices or organization of the files to be made without requiring changes in the conceptual view (or) any of the external views and hence in the application programs using the database. Thus the files may migrate from one type of physical media to another of the file structure may change without any need for changes in the application programs.

  Example, providing an access path to improve retrieval speed of section records by semester and year should not require a query to be changed, although it should become more efficient by utilizing the access path.

Definition: Physical Data independence

It is the capacity to change the Internal Schema without having to change conceptual schema.

2. Logical Data Independence

- The ability to change the conceptual schema without having to change the external schemas or application programs. When data is added or removed, only the view definition and the mappings need to be changed in the DBMS that support logical data independence.
- If the conceptual schema undergoes a logical reorganization, application programs that reference the external schema constructs must work as before.
- Logical data independence indicates that the conceptual schema can be changed without affecting the existing external schemas.
- ***Data independence*** is advantageous in the database environment since it allows for changes at one level of the database without affecting other levels. ***Logical data independence*** is more difficult to achieve than ***Physical data independence***. Since application programs are heavily dependent on the logical structure of the data they access.

Definition: Logical Data Independence

It is the capacity to change the conceptual schema without having to change in external schema or application schema.

1.19 Database Languages

A DBMS supports a variety of users and must provide appropriate languages and interfaces for each category of users.

1. ***DDL (Data Definition Language):*** used (by the DBA and/or database designers) to specify the conceptual schema.
   - ***Example:*** CREATE, DROP, TRUNCATE and ALTER statements.
     - ***SDL (Storage Definition Language)***: used for specifying the internal schema
     - ***VDL (View Definition Language)***: used for specifying the external schemas (i.e., user views)
2. ***DML (Data Manipulation Language):*** used for performing operations such as retrieval and update upon the populated database.
   - ***Example:*** SELECT, DELETE, INSERT and UPDATE statements.
3. ***TCL (Transaction Control Language):*** It is used to manage different transactions occurring within a database.
   - **Example:** COMMIT ROLLBACK and SAVEPOINT statements.
4. ***DCL (Data Control Language):*** It is used to create roles, permissions, and referential integrity as well it is used to control access to database by securing it.
   - ***Example:*** GRANT and REVOKE statements.

The above description represents some kind of ideal. In real-life, at least so far, the de facto standard DBMS language is SQL (Standard Query Language), which has constructs to support the functions needed by DDL, VDL, and DML languages. (Early versions of SQL had features in support of SDL functions, but no more.)

1.19.1 DDL (Data Definition Language)

DDL define the logical schema (relations, views etc) and storage schema stored in a Data Dictionary. In DBMS where a clear separation is maintained between the conceptual and internal levels, the Data Definition Language (DDL) is used to specify the conceptual schema only. The DDL is used to create and destroy databases and database objects. These commands will primarily be used by database administrators during the setup and removal phases of a database project. Another language, the storage definition language (SDL), is used to specify the internal schema. The mappings between the two schemas may be specified in either one of these languages. For true three-schema architecture, we would need a third language, the view definition language (VDL), to specify user views and their mappings to the conceptual schema, but in most DBMS the DDL is used to define both conceptual and external schemas.

? What is DDL?

DDL is used by the DBA and by database designer to define schemas. The DBMS will have a DDL compiler, whose function is to process DDL statements in order to identify descriptions of the schema constructs and to store the schema description in the DBMS catalogue.

1.19.2 DML (Data Manipulation Language)

DML are other set of commands in databases that lets you work with the real data items that you are concerned with. SQL commands such as SELECT, UPDATE, INSERT are used to retrieve data from database, modify data items (or do some correction or update) and insert new records inside database respectively. By this you can see DML are related with the data and not with structure of database.

? What is DML?

DML stands for Data Manipulation Language. Once database is created or defined, and the database is populated with data, user must have some means to manipulate the database by using DML. (OR)

DML is a language that enable users to access or manipulate data, as organized by the appropriate data model. Data model involves retrieval of data from the database, Insertion of new data into the database, deletion of modification of existing data.

1.19.3 DCL (Data Control Language)

Database stores huge amount of data and data are most important and sensitive to any organization. Nobody likes to grant whole access to everything inside database. For example an employee in Admin department has nothing to do with records related to Finance of the organization. So, it would be wise to restrict the employee to use only the required tables and data. For this purpose, database provides a set of commands that can be used to create new user accounts for the database, assign passwords and privileges. Each user is identified through his/her user ID and is allowed to perform certain actions only based on what privileges are granted to that user. SQL provides commands like GRANT, REVOKE commands to assign and remove user rights to access database.

? What is DCL?

DCL is used to control access to data stored in a database. DCL statements executed based on the type of action to be granted (or) revoked. Grant to allow specified users to perform specified tasks and revoke to cancel previously granted permission.

1.19.4 TCL (Transaction Control Language)

Transaction Control (TCL) statements are used to manage the changes made by DML statements. It allows statements to be grouped together into logical transactions. A transaction is a logical unit of work. All changes made to the database can be referred to as a transaction. Transaction changes can be made permanent to the database only if they are committed a transaction begins with an executable SQL statement and ends explicitly with either role back or commit statement. SQL provides commands like COMMIT, SAVEPOINT, ROLLBACK.

? What is TCL?

Statements which are used to manage the changes made by DML statements are called Transaction Control Language. It is used to manage transactions in a database. It allows statements to be grouped together into logical transactions.

1.20 DBMS Interfaces

Interfaces are the programs which convert system language to user understandable language and user language to system understandable language. So its work likes a translator which provides an environment in which a user can easily operate the system.

Types of interfaces provided by the DBMS include

1. Menu-based interfaces for web clients or browsing
2. Forms-based interfaces
3. GUI's
4. Natural Language Interfaces
5. Interfaces for parametric users
6. Interfaces for the DBA

1. Menu-Based Interfaces for Web Clients (or) Browsing

- These interfaces present the user with lists of options (called menus) that lead the user through the formation of a request.
- The main advantage of using menus is that they removes the tension of remembering specific commands and syntax of any query language.
- The query is basically composed step by step by collecting or picking options from a menu that is shown by the system. Pull-down menus are a very popular technique in Web based interfaces.
- They are also often used in browsing interface which allow a user to look through the contents of a database in an exploratory and unstructured manner.

2. Forms-Based Interfaces

- A forms-based interface displays a form to each user. Users can fill out all of the form entries to insert new data, or they can fill out only certain entries, in which case the DBMS will redeem same type of data for other remaining entries.
- These types of forms are usually designed or created and programmed for the users that have no expertise in operating system. Many DBMSs have forms specification languages which are special languages that help specify such forms.

Example: SQL* Forms is a form-based language that specifies queries using a form designed in conjunction with the relational database schema.

3. Graphical User Interfaces (GUI's)

- A GUI typically displays a schema to the user in diagrammatic form. The user then can specify a query by manipulating the diagram. In many cases, GUIs utilize both menus and forms.

- Most GUIs use a pointing device such as mouse, to pick a certain part of the displayed schema diagram.

4. **Natural Language Interfaces**
   - These interfaces accept request written in English or some other language and attempt to understand them. A Natural language interface has its own schema, which is similar to the database conceptual schema as well as a dictionary of important words.
   - The natural language interface refers to the words in its schema as well as to the set of standard words in a dictionary to interpret the request. If the interpretation is successful, the interface generates a high-level query corresponding to the natural language and submits it to the DBMS for processing, otherwise a dialogue is started with the user to clarify any provided condition or request.
   - The main disadvantage with this is that the capabilities of this type of interfaces are not that much advance.
5. **Interfaces for Parametric Users**
   - Parametric users have small set of operations they perform.
   - Analysts and programmers design and implement a special interface for each class of naïve users.
   - Often a small set of commands included to minimize the number of keystrokes required. (I.e. function keys)
6. **Interfaces for the DBA**
   - Most database system contains privileged commands that can be used only by the DBA's staff. These include commands for creating accounts, setting system parameters, granting account authorization, changing a schema, reorganizing the storage structures of a databases.

1.21 Classifications of DBMS

The DBMS can be classified into different categories on the basis of several criteria such as the data model they are using, number of users they support, number of sites over which the database is distributed and the purpose they serve.

1. **Classification based on Data Model**
   - Relational data model
   - Network data model
   - Object -Relational data model
   - Hierarchical data model
   - Object-Oriented data model
2. **Classification based on Number of Users**
   - **Single User Systems**- single-user system the database resides on one computer and is only accessed by one user at a time.
   - **Multi User Systems** - multiple users can access the database simultaneously. In multi-user DBMS, the data is both integrated and shared. For example, the Online Book database is a multi-user database system in which the data of books, authors, and publishers are stored centrally and can be accessed by many users.

3. **Classification based on Number of Sites**
   - **Centralized** – data is stored at single site.
   - **Distributes** – database and DBMS software stored over many sites connected by network
   - **Homogeneous** – use same DBMS software at multiple sites.
   - **Online Transaction Processing (OLTP)** - used for data entry and retrieval. It is used in various industries, such as banking, airlines, supermarkets, manufacturing, etc.
4. **Classification based on Cost**
   - Low-end systems
   - High-end systems,
5. **Classification based on Purpose**
   - General purpose
   - Special purpose

   DBMS is a general purpose software system. It can be designed for specific purposes such as airline or railway reservation. Such systems cannot be used for other applications without major changes.

1.22 Review Questions

I. Two Marks Questions and Answers

1. What is Data and Information? Give an example

Ans. Data is a representation of a Facts, figures, statistics etc. having no particular meaning. Data can be in the form of numbers, characters, symbols, or even pictures.

Example: 1, Computer, Student, Rajesh, 3.14 etc

It is a processed data (or) Collection of data which is having perfect meaning is called information.

Example: Student's Roll No is 19

2. Define field and record with an example.

Ans. A field is a single piece of information; a record is one complete set of fields; and a file is a collection of records.

Example: A telephone book is analogous to a file. It contains a list of records, each of which consists of three fields: name, address, and telephone number.

3. What is database and DBMS?

Ans. **Database:** A database system is a way of organizing information on a computer, implemented by a set of computer programs.

DBMS: A database management system is a software system. It allows access to the data in a database. It consists of a set of interrelated data together with a set of programs to access those data.

4. List advantages of DBMS?

Ans. Advantages of DBMS:

- Reduced Data Redundancy
- Providing Backup and Recovery
- Restricting Unauthorized Access
- Providing Multiple User Interfaces

5. List disadvantages of DBMS?

Ans. Disadvantages of DBMS

- Cost of Data Conversion
- Cost of Staff Training
- Cost of Hardware & Software
- Appointing Technical Staff

6. Write few examples of traditional database application?

Ans. Examples of traditional database system

- Library Management System
- Payroll Management System
- Inventory Management System
- Airline Reservation

7. Write few examples of advanced database application?

Ans. Examples of advanced database system

- Multimedia Database
- Web Database
- Mobile Database
- Real Time Database

8. What is DBA and Database Administration?

Ans. DBA: The person who is responsible for controlling and coordinating the DBMS system is called Database System Administrator (DBA).

Database Administration: Database administration is more of an operational or technical level function responsible for physical database design, security enforcement, and database performance. Tasks include maintaining the data dictionary, monitoring performance, and enforcing organizational standards and security.

9. What is Data Model? And name different type of data models.

Ans. Data Model can be defined as an integrated collection of concepts for describing and manipulating data, relationships between data, and constraints on the data in an organization.

Types of data models: There are five models of DBMS, which are distinguished based on how they represent the data contained:

- Relational data model
- Hierarchical data model
- Network data model
- Object-Oriented data model
- Object -Relational data model

10. What is Relational and Network Data Model?

Ans. Relational Data Model: A Relational model database is defined as a database that allows to group its data items into one or more independent tables that can be related to one another by using common fields related to each table.

Network Data Model: Network model organizes data using two fundamental constructs called records and sets. Records contain fields, Sets define one-to-many relationships between records.

11. Define Hierarchical and Object Oriented Data Model?

Ans. Hierarchical Data Model: The hierarchical data model organizes data in a tree structure. There is a hierarchy of parent and child data segments.

Object Oriented Data Model: A data model is a logical organization of the real world objects (entities), constraints on them and relationship among objects. A data model that captures is an object-oriented data model. A object - oriented database is a collection of objects whose behavior, state and the relationships are called object-oriented data model.

12. Explain Object Relational Data Model?

Ans. An object-relational database (ORD) is a database management system (DBMS) that's composed of both a relational database (RDBMS) and an object-oriented database (OODBMS). ORD supports the basic components of any object-oriented database model in its schemas and the query language used, such as objects, classes and inheritance.

13. What is schema and database instance?

Ans. **Schema:** A database schema is the logical representation of a database, which shows how the data is stored logically in the the database. It contains list of attributes and instruction that informs the database engine that how the data is organized and how the elements are related to each other.

Database Instance: The data in the database at a particular moment in time is called the Database state (or) Snapshot (or) instance.

14. What is mapping and data independence?

Ans. **Mapping:** The processes of transforming requests and results between different levels is called mappings.

Data independence: The ability to modify a schema definition in one level without affecting schema definition in the next higher level is called "Data Independence".

15. What is DDL and DML?

Ans. **DDL:** DDL is used by the DBA and by database designer to define both schemas. The DBMS will have a DDL compiler, whose function is to process DDL statements in order to identify descriptions of the schema constructs and to store the schema description in the DBMS catalogue.

DML: DML stands for Data Manipulation Language. Once database is created or defined, and the database is populated with data, user must have some means to manipulate the database by using DML.

(OR)

DML is a language that enable users to access or manipulate data, as organized by the appropriate data model. Data model involves retrieval of data from the database, Insertion of new data into the database, deletion of modification of existing data.

16. Define DCL and TCL.

Ans. **DCL:** DCL is used to control access to data stored in a database. DCL statements executed based on the type of action to be granted (or) revoked. Grant to allow specified users to perform specified tasks and revoke to cancel previously granted permission.

TCL: Statements which are used to manage the changes made by DML statements are called Transaction Control Language. Used to manage transactions in a database. It allows statements to be grouped together into logical transactions.

17. What is distributed and centalized DBMS.

Ans. **Distributed DBMS:** A distributed database is a database in which all storage devices are not attached to a common CPU. Data may be stored in multiple sites separate from each other.

Centralized DBMS: A centralized DBMS in which all the DBMS functionality, application program execution and user interface processing were carried out on a single machine.

18. Mention the three layers of DBMS Architecture.

Ans. (a) External Level
(b) Conceptual Level
(c) Internal Level

II. Five Marks Questions

1. Explain traditional file oriented approach.
2. How database approaches differ from traditional file system concepts?
3. Explain the disadvantages of traditional file oriented approach?
4. Explain the functions of DBMS?
5. Write few applications of DBMS?
6. Define DBA? Explain the role and responsibilities of DBA.
7. Explain the functions of DBA.
8. When not to use a DBMS?
9. Describe any two types of data models with an example.
10. Explain relational model with advantages and disadvantages?
11. Describe object oriented data model?
12. What object oriented relational model?
13. Describe the three schema architecture of DBMS with a neat diagram.
14. What mapping? Explain two types of mapping.
15. What is data independence? Explain the two types of data independence
16. Differentiate between logical data independence and physical data independence.
17. Give the classification of DBMS.

III. Eight Marks Questions

1. Explain the characteristics of database approach.
2. What the advantages and disadvantages of DBMS.
3. Describe different users behind DBMS.
4. Explain a simplified database system in detail with a neat diagram.
5. Write a short notes on history of database application.
6. Describe the different data models with advantages and disadvantages. Explain one example for each.
7. Write a short note on database languages.
8. Explain the different DBMS interfaces.
9. Write a short note on distributed and centralized DBMS architecture.

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